

Guide for trident v2.2.2.0 and higher

Contents

1	Installation	2
2	Overview	2
2.1	Logging and command line output	4
2.2	Package duplicates and versions	4
2.3	Individual/Sample duplicates	5
2.4	Group names in .fam files	5
2.5	Whitespaces in the .janno file	5
3	Init command	5
4	Fetch command	7
5	Forge command	8
5.1	The forge selection language	12
5.1.1	Inclusion queries	13
5.1.2	Exclusion queries	14
5.2	Ordered output	14
5.2.1	Reordering samples in a package	14
5.3	Treatment of the genotype data while merging	15
5.3.1	Strand flips	15
5.4	Treatment of the .janno file while merging	16
5.5	Treatment of the .ssf file while merging	16
5.6	Treatment of the .bib file while merging	17
5.7	Output modes	17
5.8	Other options	18
6	Genoconvert command	18
7	Jannocoalesce command	20
8	Rectify command	22
9	List command	24
10	Summarise command	25
11	Survey command	26

33 1 Installation

34 See the Poseidon website (<https://www.poseidon-adna.org/#/trident>) or the GitHub repository (<https://github.com/poseidon-framework/poseidon-hs>) for up-to-date installation instructions.

36 2 Overview

37 Trident is a command line software tool structured in multiple subcommands. If you installed it properly you
38 can call it on the command line by typing `trident`. This will show an overview of the general options and all
39 subcommands, which are explained in detail below.

```
Usage: trident [--version] [--logMode MODE | --debug | --noWarn]
          [--errLength INT] [--inPlinkPopName MODE] (COMMAND | COMMAND)
```

trident is a management and analysis tool for Poseidon packages. Report issues
here: <https://github.com/poseidon-framework/poseidon-hs/issues>

Available options:

<code>-h, --help</code>	Show this help text
<code>--version</code>	Show version number
<code>--logMode MODE</code>	How information should be reported: NoLog, SimpleLog, DefaultLog, ServerLog or VerboseLog. (default: DefaultLog)
<code>--debug</code>	Short for <code>--logMode VerboseLog</code> .
<code>--noWarn</code>	Short for <code>--logMode NoWarnLog</code> .
<code>--errLength INT</code>	After how many characters should a potential genotype data parsing error message be truncated. "Inf" for no truncation. (default: CharCount 1500)
<code>--inPlinkPopName MODE</code>	Where to read the population/group name from the FAM file in Plink-format. Three options are possible: asFamily (default) asPhenotype asBoth.

Package creation and manipulation commands:

<code>init</code>	Create a new Poseidon package from genotype data
<code>fetch</code>	Download data from a remote Poseidon repository
<code>forge</code>	Select packages, groups or individuals and create a new Poseidon package from them
<code>genoconvert</code>	Convert the genotype data in a Poseidon package to a different file format
<code>jannocoalesce</code>	Coalesce information from one or multiple janno files to another one
<code>rectify</code>	Adjust POSEIDON.yml files automatically to package changes

Inspection commands:

<code>list</code>	List packages, groups or individuals from local or remote Poseidon repositories
<code>summarise</code>	Get an overview over the content of one or multiple Poseidon packages
<code>survey</code>	Survey the degree of context information completeness for Poseidon packages
<code>validate</code>	Check Poseidon packages or package components for structural correctness

40 `trident` allows to work directly with genotype data (see `-p` below), but it is optimized for the interaction
41 with Poseidon packages, which wrap and contextualize the data. Most `trident` subcommands therefore have a
42 central parameter, called `--baseDir` or simply `-d` to specify one or more base directories to look for packages.
43 For example, if all Poseidon packages live inside a repository at `/path/to/poseidon/packages` you would
44 simply say `trident <subcommand> -d /path/to/poseidon/dirs/` and `trident` would automatically search
45 all subdirectories inside of the repository for valid Poseidon packages (as identified by valid `POSEIDON.yml`
46 files).

47 You can arrange a Poseidon repository in a hierarchical way. For example:

```
/path/to/poseidon/packages
  /modern
    /2019_poseidon_package1
    /2019_poseidon_package2
  /ancient
    /...
    /...
  /Reference_Genomes
    /...
    /...
```

48 This structure then allows to select only the level of packages you are interested in, even individual ones. `-d`
49 can be given multiple times, which is particularly useful as you may have your own data to co-analyse with
50 external reference data. In this case you simply need to provide your own genotype data as yet another Poseidon
51 package to be added to your `trident` command. For example, you may have genotype data in `EIGENSTRAT`
52 format (`trident` supports `EIGENSTRAT`, `PLINK` and `VCF` as formats):

```
~/my_project/my_project.geno
~/my_project/my_project.snp
~/my_project/my_project.ind
```

53 Then you can transform that into a skeleton Poseidon package with the `init` command. You can also do it
54 manually by simply adding a `POSEIDON.yml` file, with, for example, the following content:

```
poseidonVersion: 3.0.0
title: My_awesome_project
description: Unpublished genetic data from my awesome project
contributor:
  - name: Stephan Schiffels
```

```

    email: schiffels@institute.org
packageVersion: 0.1.0
lastModified: 2020-10-07
genotypeData:
  format: EIGENSTRAT
  genoFile: my_project.geno
  snpFile: my_project.snp
  indFile: my_project.ind
  snpSet: 1240K
  referenceGenomeAssembly: GRCh37
  referenceGenomeAssemblyURL: https://www.ncbi.nlm.nih.gov/datasets/genome/GCA_000001405.14
jannoFile: my_project.janno
bibFile: sources.bib

```

Two remarks: 1) All file paths in this POSEIDON.yml file are considered *relative* to the directory in which POSEIDON.yml resides. For this example we assume that this file is added into the same directory as the three genotype files. 2) Besides the genotype data files there are two (technically optional) files referenced in this example: `sources.bib` and `my_project.janno`. Of course you can add them manually - `init` automatically creates empty dummy versions.

Once you have set up your own Poseidon package (which is really only a skeleton so far), you can add it to your `trident` analysis, by simply adding your project directory to the command using `-d`, for example:

```

trident list -d /path/to/poseidon/packages/modern \
-d /path/to/poseidon/packages/ReferenceGenomes \
-d ~/my_project \
--packages

```

2.1 Logging and command line output

For all subcommands the general argument `--logMode` defines how `trident` reports messages (to stderr) on the command line:

- *NoLog*: Hides all messages.
- *SimpleLog*: Plain and simple output.
- *DefaultLog*: Adds the severity indicators (log levels) **Info**, **Warning** and **Error** before each message. This is the default setting.
- *NoWarnLog*: As *DefaultLog*, but omits messages on the **Warning** log level.
- *ServerLog*: Additionally adds timestamps before each message.
- *VerboseLog*: Shows not just messages on the log levels **Info**, **Warning** and **Error** like the other modes, but also on the more verbose level **Debug**. Use this mostly relevant for debugging.

`--debug` is short for `--logMode VerboseLog` to activate this important log level more easily. And `--noWarn` is short for `--logMode NoWarnLog`.

2.2 Package duplicates and versions

- For `trident` multiple packages in a set of base directories can share the same **title**, if they have different **packageVersion** numbers. If the version numbers are also identical or missing, then `trident` stops with

an exception.

- The **trident** subcommands **genoconvert**, **list**, **rectify**, **survey** and **validate** by default consider all versions of each Poseidon package in the given base directories. The **--onlyLatest** flag causes them to instead only consider the latest versions.
- **fetch** and **forge** generally consider all package versions. Their selection language (see below) allows for detailed version handling.
- **summarize** and **jannocoalesce** consider always only the latest package versions.

2.3 Individual/Sample duplicates

- **Poseidon_IDs** (so individual/sample names) within one package have to be unique, or **trident** will stop.
- We also discourage sample duplicates across packages in package repositories, but **trident** will generally continue with them. **validate** will fail though, if the **--ignoreDuplicates** flag is not set.
- **forge** offers a special mechanism to resolve sample duplicates within its selection language.

2.4 Group names in .fam files

The **.fam** file of PLINK-formatted genotype data is used inconsistently across different popular aDNA software tools to store group/population name information. The (global) **trident** option **--inPlinkPopName** with the arguments **asFamily** (default), **asPhenotype** and **asBoth** allows to control the reading of the population name from PLINK **.fam** files. The subcommands that write genotype data (**forge**, **genoconvert**) have a corresponding option **--outPlinkPopName** to specify this for the output.

2.5 Whitespaces in the .janno file

While reading the **.janno** file **trident** trims all leading and trailing whitespaces around individual cells. Also all instances of the **No-Break Space** unicode character will be removed. This means these whitespaces will not be preserved when a package is forged.

3 Init command

init creates a new Poseidon package from genotype data files. It adds a **POSEIDON.yml** file, a dummy **.janno** file for context information and an empty **.bib** file for literature references.

Command line details

```
Usage: trident init ((-p|--genoOne FILE) | --genoFile FILE --snpFile FILE
                    --indFile FILE |
                    --bedFile FILE --bimFile FILE --famFile FILE |
                    --vcfFile FILE) [--snpSet SET] (-o|--outPackagePath DIR)
                    [-n|--outPackageName STRING] [--minimal]
```

Create a new Poseidon package from genotype data

Available options:

-h,--help	Show this help text
-p,--genoOne FILE	One of the input genotype data files. Expects .bed , .bed.gz , .bim , .bim.gz or .fam for PLINK, .geno ,

.geno.gz, .snp, .snp.gz or .ind for EIGENSTRAT, or.vcf or .vcf.gz for VCF. In case of EIGENSTRAT and PLINK, the two other files must be in the same directory and must have the same base name. If a gzipped file is given, it is assumed that the file pairs (.geno.gz, .snp.gz) or (.bim.gz, .bed.gz) are both zipped, but not the .fam or .ind file. If a .ind or .fam file is given, it is assumed that none of the file triples is zipped.

`--genoFile FILE` Eigenstrat genotype matrix, optionally gzipped. Accepted file endings are .geno, .geno.gz

`--snpFile FILE` Eigenstrat snp positions file, optionally gzipped. Accepted file endings are .snp, .snp.gz

`--indFile FILE` Eigenstrat individual file. Accepted file endings are .ind

`--bedFile FILE` Plink genotype matrix, optionally gzipped. Accepted file endings are .bed, .bed.gz

`--bimFile FILE` Plink snp positions file, optionally gzipped. Accepted file endings are .bim, .bim.gz

`--famFile FILE` Plink individual file. Accepted file endings are .fam

`--vcfFile FILE` VCF (Variant Call Format) file, optionally gzipped. Accepted file endings are .vcf, .vcf.gz

`--snpSet SET` The snpSet of the package: 1240K, HumanOrigins or Other. Only relevant for data input with `-p|--genoOne` or `--genoFile + --snpFile + --indFile`, because the packages in a `-d|--baseDir` already have this information in their respective POSEIDON.yml files. (default: Other)

`-o,--outPackagePath DIR` Path to the output package directory.

`-n,--outPackageName STRING` The output package name. This is optional: If no name is provided, then the package name defaults to the basename of the (mandatory) `--outPackagePath` argument. (default: Nothing)

`--minimal` Should the output Poseidon package be reduced to a necessary minimum?

104 The command

```
trident init \
  --genoFile path/to/genoFile.geno \
  --snpFile path/to/snpFile.snp \
  --indFile path/to/indFile.ind \
  --snpSet 1240K|HumanOrigins|Other \
  -o path/to/new_package_name
```

requires the paths to the respective files (`--genoFile --snpFile --indFile | --bedFile --bimFile --famFile | --vcfFile`), and optionally the “shape” of these files (`--snpSet`), so if they cover the 1240K, the HumanOrigins or an Other SNP set.

A simpler interface is available with `-p` (+ `--snpSet`), which only requires a path to one of the genotype data files and automatically discovers the others if they share the same base name:

```
trident init \
  -p path/to/genoFile \
  --snpSet 1240K|HumanOrigins|Other \
  -o path/to/new_package_name
```

The following file extensions are expected:

	EIGENSTRAT	PLINK	VCF
genoFile	.geno	.bed	.vcf
snpFile	.snp	.bim	—
indFile	.ind	.fam	—

The output package created by `init` is located in a new directory `-o`, which should not already exist when `init` is called, and gets the package title corresponding to the basename of `-o`. You can also set the title explicitly with `-n`.

The `--minimal` flag causes `init` to create a minimal package with a very basic POSEIDON.yml and no `.bib` and `.janno` files.

4 Fetch command

`fetch` allows to download Poseidon packages from a remote Poseidon server via a Web API. This server provides all packages in the Poseidon public archives.

Command line details

```
Usage: trident fetch (-d|--baseDir DIR)
        (--downloadAll |
        (--fetchFile FILE | (-f|--fetchString DSL)))
        [--remoteURL URL] [--archive STRING]
```

Download data from a remote Poseidon repository

Available options:

<code>-h,--help</code>	Show this help text
<code>-d,--baseDir DIR</code>	A base directory to search for Poseidon packages.
<code>--downloadAll</code>	Download all packages the server is offering.
<code>--fetchFile FILE</code>	A file with a list of packages. Works just as <code>-f</code> , but multiple values can also be separated by newline, not just by comma. <code>-f</code> and <code>--fetchFile</code> can be combined.
<code>-f,--fetchString DSL</code>	List of packages to be downloaded from the remote

```

server. Package names should be wrapped in asterisks:
*package_title*. You can combine multiple values with
comma, so for example: "*package_1*, *package_2*,
*package_3*". fetchString uses the same parser as
forgeString, but does not allow excludes. If groups
or individuals are specified, then packages which
include these groups or individuals are included in
the download.

--remoteURL URL          URL of the remote Poseidon server.
                          (default: "https://server.poseidon-adna.org")

--archive STRING         The name of the Poseidon package archive that should
                          be queried. If not given, then the query falls back
                          to the default archive of the server selected with
                          --remoteURL. See the archive documentation at
                          https://www.poseidon-adna.org/#/archive_overview for
                          a list of archives currently available from the
                          official Poseidon Web API. (default: Nothing)

```

120 It works with

```

trident fetch -d ... -d ... \
  -f "*package_title_1*,*package_title_2-1.0.1*,group_name,<individual1>"

```

121 and the entities you want to download must be listed either in a simple string of comma-separated values, which
 122 can be passed via `-f/--fetchString`, or in a text file (`--fetchFile`). Entities are then combined from these
 123 sources.

124 Entities are specified using a special syntax (see also the documentation of `forge` below): packages are
 125 wrapped in asterisks, with or without a version number appended after a dash (e.g. `*package_title*` or
 126 `*package_title-1.2.3`), group names are spelled as is, and individual names are wrapped in angular brackets
 127 (e.g. `<individual1>`). Fetch will figure out which packages need to be downloaded to include all specified entities.

128 `--downloadAll`, which can be given instead of `-f` and `--fetchFile`, causes fetch to download all packages from
 129 the server. The downloaded packages are added in the first (!) `-d` directory (which gets created if it doesn't
 130 exist), but downloads are only performed if the respective packages are not already present in the latest version
 131 in any of the `-d` directories.

132 Note that `trident fetch` is usually used in a workflow with `trident list --remote`: First one inspects what
 133 is available on the server with `list`, to then compile a custom, targeted `fetch` command.

134 `fetch` has the optional arguments `--remote https://...` to name an alternative Poseidon server and
 135 `--archive` to select a specific Poseidon archive on the server.

136 5 Forge command

137 `forge` creates new Poseidon packages by extracting and merging packages, populations and individuals/samples
 138 from Poseidon repositories.

139 Command line details

```
Usage: trident forge ((-d|--baseDir DIR) |
                    ((-p|--genoOne FILE) | --genoFile FILE --snpFile FILE
                     --indFile FILE |
                     --bedFile FILE --bimFile FILE --famFile FILE |
                     --vcfFile FILE) [--snpSet SET])
[--forgeFile FILE | (-f|--forgeString DSL)]
[--selectSnps FILE] [--intersect] [--strandCheck]
[--skipIncongruentSNPs] [--outFormat FORMAT]
[--onlyGeno | --minimal | --preserve] [-z|--zip]
(-o|--outPackagePath DIR) [-n|--outPackageName STRING]
[--packagewise] [--outPlinkPopName MODE] [--ordered]
[--addTrace]
```

Select packages, groups or individuals and create a new Poseidon package from them

Available options:

<code>-h,--help</code>	Show this help text
<code>-d,--baseDir DIR</code>	A base directory to search for Poseidon packages.
<code>-p,--genoOne FILE</code>	One of the input genotype data files. Expects .bed, .bed.gz, .bim, .bim.gz or .fam for PLINK, .geno, .geno.gz, .snp, .snp.gz or .ind for EIGENSTRAT, or.vcf or .vcf.gz for VCF. In case of EIGENSTRAT and PLINK, the two other files must be in the same directory and must have the same base name. If a gzipped file is given, it is assumed that the file pairs (.geno.gz, .snp.gz) or (.bim.gz, .bed.gz) are both zipped, but not the .fam or .ind file. If a .ind or .fam file is given, it is assumed that none of the file triples is zipped.
<code>--genoFile FILE</code>	Eigenstrat genotype matrix, optionally gzipped. Accepted file endings are .geno, .geno.gz
<code>--snpFile FILE</code>	Eigenstrat snp positions file, optionally gzipped. Accepted file endings are .snp, .snp.gz
<code>--indFile FILE</code>	Eigenstrat individual file. Accepted file endings are .ind
<code>--bedFile FILE</code>	Plink genotype matrix, optionally gzipped. Accepted file endings are .bed, .bed.gz
<code>--bimFile FILE</code>	Plink snp positions file, optionally gzipped. Accepted file endings are .bim, .bim.gz
<code>--famFile FILE</code>	Plink individual file. Accepted file endings are .fam
<code>--vcfFile FILE</code>	VCF (Variant Call Format) file, optional all gzipped. Accepted file endings are .vcf, .vcf.gz
<code>--snpSet SET</code>	The snpSet of the package: 1240K, HumanOrigins or Other. Only relevant for data input with -p --genoOne

	or <code>--genoFile + --snpFile + --indFile</code>, because the packages in a <code>-d --baseDir</code> already have this information in their respective POSEIDON.yml files. (default: Other)
<code>--forgeFile FILE</code>	A file with a list of packages, groups or individual samples. Works just as <code>-f</code>, but multiple values can also be separated by newline, not just by comma. Empty lines are ignored and comments start with <code>"#"</code>, so everything after <code>"#"</code> is ignored in one line. Multiple instances of <code>-f</code> and <code>--forgeFile</code> can be given. They will be evaluated according to their input order on the command line.
<code>-f,--forgeString DSL</code>	List of packages, groups or individual samples to be combined in the output package. Packages follow the syntax <code>*package_title*</code>, populations/groups are simply <code>group_id</code> and individuals <code><individual_id></code>. You can combine multiple values with comma, so for example: <code>"*package_1*, <individual_1>, <individual_2>, group_1"</code>. Duplicates are treated as one entry. Negative selection is possible by prepending <code>"-"</code> to the entity you want to exclude (e.g. <code>"*package_1*, -<individual_1>, -group_1"</code>). <code>forge</code> will apply excludes and includes in order. If the first entity is negative, then <code>forge</code> will assume you want to merge all individuals in the packages found in the <code>baseDirs</code> (except the ones explicitly excluded) before the exclude entities are applied. An empty <code>forgeString</code> (and no <code>--forgeFile</code>) will therefore merge all available individuals. If there are individuals in your input packages with equal individual id, but different main group or source package, they can be specified with the special syntax <code>"<package:group:individual>"</code>.
<code>--selectSnps FILE</code>	To extract specific SNPs during this <code>forge</code> operation, provide a Snp file. Can be either Eigenstrat (file ending must be <code>'.snp'</code> or <code>'.snp.gz'</code>) or Plink (file ending must be <code>'.bim'</code> or <code>'.bim.gz'</code>). When this option is set, the output package will have exactly the SNPs listed in this file. Any SNP not listed in the file will be excluded. If option <code>'--intersect'</code> is also set, only the SNPs overlapping between the SNP file and the forged packages are output. (default: Nothing)
<code>--intersect</code>	Whether to output the intersection of the genotype files to be forged. The default (if this option is

not set) is to output the union of all SNPs, with genotypes defined as missing in those packages which do not have a SNP that is present in another package. With this option set, the forged dataset will typically have fewer SNPs, but less missingness.

`--strandCheck` Whether to allow strand flips in the genotype data. Note that this will remove any A/T and G/C SNPs from the data, as for those we cannot determine the correct strand orientation.

`--skipIncongruentSNPs` Whether to skip SNPs with incongruent alleles.

`--outFormat FORMAT` The format of the output genotype data: EIGENSTRAT, PLINK or VCF. (default: PLINK)

`--onlyGeno` Should only the resulting genotype data be returned? This means the output will not be a Poseidon package.

`--minimal` Should the output Poseidon package be reduced to a necessary minimum?

`--preserve` Should the output Poseidon package mimic the input package? Only works for a singular source package! With this option some fields of the source package's POSEIDON.yml file, its README file and its CHANGELOG file (if available) are copied to the output package. The order of .janno file columns and .bib file entries are also preserved.

`-z,--zip` Should the resulting genotype- and snp-files be gzipped?

`-o,--outPackagePath DIR` Path to the output package directory.

`-n,--outPackageName STRING` The output package name. This is optional: If no name is provided, then the package name defaults to the basename of the (mandatory) `--outPackagePath` argument. (default: Nothing)

`--packagewise` Skip the within-package selection step in forge. This will result in outputting all individuals in the relevant packages, and hence a superset of the requested individuals/groups. It may result in better performance in cases where one wants to forge entire packages or almost entire packages. Details: Forge conceptually performs two types of selection: First, it identifies which packages in the supplied base directories are relevant to the requested forge, i.e. whether they are either explicitly listed using `*PackageName*`, or because they contain selected individuals or groups. Second, within each relevant package, individuals which are not requested are removed. This option skips only the second step, but

```

--outPlinkPopName MODE   still performs the first.
                          Where to write the population/group name into the FAM
                          file in Plink-format. Three options are possible:
                          asFamily (default) | asPhenotype | asBoth. See also
                          --inPlinkPopName.

--ordered                With this option, the output of forge is ordered
                          according to the entities given.

--addTrace               This option will add a field named Source_Package to
                          the .janno file, containing the source package from
                          which a given sample was derived. If the field
                          already exists, it will be extended with the new
                          source package using semicolons as separators.

```

140 `forge` can be used with

```

trident forge -d ... -d ... \
  -f "*package_name*, group_id, <individual_id>" \
  -o path/to/new_package_name

```

141 where the entities (packages, groups/populations, individuals/samples) you want in the output package can be
 142 denoted either as a string on the command line (`-f/--forgeString`), or in an input text file (`--forgeFile`).
 143 See the section below for the syntax of this selection language. Do not forget to wrap the `--forgeString` query
 144 in quotes.

145 Including one or multiple Poseidon packages with `-d` is not the only way to include data for a forge operation. It is
 146 also possible to consider unpackaged genotype data directly with `-p (+ --snpSet)`, `--genoFile + --snpFile`
 147 `+ --indFile (+ --snpSet)` (for EIGENSTRAT data), `--bedFile + --bimFile + --famFile (+ --snpSet)`
 148 (for PLINK data) or `--vcfFile (+ --snpSet)` (for VCF data). This makes the following example possible, where
 149 we merge data from one Poseidon package and two unpackaged genotype datasets to get a new EIGENSTRAT
 150 dataset.

```

trident forge \
  -d 2017_GonzalesFortesCurrentBiology \
  -p 2018_VeeramahPNAS/2018_VeeramahPNAS.fam \
  --bedFile 2017_HaberAJHG/2017_HaberAJHG.bed \
  --bimFile 2017_HaberAJHG/2017_HaberAJHG.bim \
  --famFile 2017_HaberAJHG/2017_HaberAJHG.fam \
  -f "<STR241.SG>,<ERS1790729.SG>,Iberia_HG.SG" \
  -o testpackage \
  --outFormat EIGENSTRAT \
  --onlyGeno

```

151 5.1 The forge selection language

152 The text in `--forgeString` and `--forgeFile` (and with a reduced syntax also in `--fetchString` and
 153 `--fetchFile`) are parsed as a domain specific query language that describes precisely which entities should be
 154 compiled in the output package of a given `forge` operation. The language has multiple syntactic elements and a
 155 specific evaluation logic.

156 In general a `--forgeString` query consists of multiple entities, separated by `,`. The main entities are Poseidon
157 packages, groups/populations and individuals/samples:

- 158 • Each package title is surrounded by `*`: `*package*`. That means if you want all individuals of the Poseidon
159 package `2019_Jeong_InnerEurasia` in the output package you would add `*2019_Jeong_InnerEurasia*`
160 to the query.
- 161 • Groups/populations are not specially marked: `group`. So to get all individuals of the group
162 `Swiss_Roman_Period`, you would simply add `Swiss_Roman_Period`.
- 163 • Individuals/samples are surrounded by `<` and `>`: `<individual>`. `ALA026` therefore becomes `<ALA026>`. A sec-
164 ond way to denote individuals is with the more verbose and specific syntax `<package:group:individual>`.
165 Such defined individuals take precedence over differently defined ones (so directly with `<individual>` or
166 as a subset of `*package*` or `group`). This allows to resolve duplication issues precisely – at least in cases
167 where the duplicated individuals differ in source package or primary group.
- 168 • Package versions can be appended to package names, such as `*package-1.2.3*`.
- 169 • This also works with the verbose individual syntax: `<package-1.2.3:group:individual>`.

170 Note that this syntax introduces a number of reserved characters with special meaning (`':'`, `'`, `'<`, `'>`,
171 `'*`). If the name of a package, individual, or group contains any of these, `trident` may fail to parse the entities
172 correctly. To solve this entity names can be quoted, so wrapped in `"`s (or `'`s). That makes it possible to include
173 group names like e.g. `"Belgium_<1000BC"` or individual names like `<"Individual3,4">`. The quotes must be
174 placed directly around the name, not around the entity term. That means `<"...">` will work, but not `"<...>"`.

175 In the `--forgeFile` each line is treated as a separate `forgeString`, empty lines are ignored and `#` symbols start
176 comments. So this is a valid example of a `forgeFile`:

```
# Packages
*pac1*, *pac2-1.2.3*

# Groups and individuals from other packages beyond pac1 and pac2
group1, <ind1>, group2, <ind2>, <pac3:group3:ind3>

# pac2 has two outlier individuals that should be ignored
-<ind4>           # This one has very low coverage
-<pac2:group4:ind5> # This one is from a different time period
```

177 By prepending `-` to entities, we can exclude them from the forged package (this feature is not available for
178 `fetch`). `forge` figures out the final list of samples to include by interpreting all `forge`-entities in order. So an
179 entity list `*pac1*,-<ind1>,group1` may result in a different outcome than `*pac1*,group1,-<ind1>`, depending
180 on whether `<ind1>` belongs to `group1` or not.

181 If the `forge` entity list starts with a negative entity, or if the entity list is empty, `forge` will implicitly assume
182 you want to include all individuals in all **latest** versions of packages found in the base directories (except the
183 ones explicitly excluded, of course).

184 The specific semantics of the various ways to include or exclude entities are as follows:

185 5.1.1 Inclusion queries

- 186 • `*pac1*`: Select all individuals in the latest version of package “pac1”
- 187 • `*pac1-1.0.1*`: Select all individuals in package “pac1” with version “1.0.1”

- **group1**: Select all individuals associated with “group1” in all latest versions of all packages
- **<ind1>**: Select the individual named “ind1”, searching in all latest packages.
- **<pac1:group1:ind1>**: Select the individual named “ind1” associated with “group1” in the latest version of package “pac1”
- **<pac1-1.0.1:group1:ind1>**: Select the individual named “ind1” associated with “group1” in the package “pac1” with version “1.0.1”

5.1.2 Exclusion queries

- **-*pac1***: Remove all individuals in all versions of package “pac1”
- **-*pac1-1.0.1***: Remove only individuals in package “pac1” with version “1.0.1” (but leave other versions in)
- **-group1**: Remove all individuals associated with “group1” in all versions of all packages (not just the latest)
- **-<ind1>**: Remove all individuals named “ind1” in all versions of all packages (not just the latest)
- **-<pac1:group1:ind1>**: Remove the individual named “ind1” associated with “group1”, searching in all versions of package “pac1”
- **-<pac1-1.0.1:group1:ind1>**: Remove the individual named “ind1” associated with “group1”, but only if they are in “pac1” with version “1.0.1”

If a query results in multiple individuals with the same name, **forge** will throw an error.

5.2 Ordered output

By default the order of samples in a Poseidon package created with **forge** depends on the order in which the relevant source packages are discovered by **trident** (e.g. when it crawls for packages in the **-d** base directories) and then the sample order within these packages.

The option **--ordered** gives more control over the output order. It causes **trident** to output the resulting package with samples ordered according to the selection in **-f** or **--forgeFile**. This works through an alternative, slower sample selection algorithm that loops through the list of entities and checks for each entity which samples it adds or removes respectively to and from the final selection.

For simple, positive selection, packages, groups and samples are added as expected. Negative selection removes samples from the list again. If an entity is selected twice via positive selection, then its first occurrence is considered for the ordering.

5.2.1 Reordering samples in a package

One particular application of **--ordered** is the reordering of samples in an existing Poseidon package, here for example **MyPac**. We suggest the following workflow for this application:

1. Generate a **--forgeFile** with the desired order of the samples in **MyPac**. This can be done manually or with any suitable tool. Here is an example, where we employ **qjanno** to generate a **forge** selection so that the samples are ordered alphabetically by their **Poseidon_ID**:

```
qjanno "SELECT '<||Poseidon_ID||>' FROM d(MyPac) ORDER BY Poseidon_ID" \
--raw --noOutHeader > myOrder.txt
```

2. Use **trident forge** with **--ordered** and **--preserve** (see below) to create the package with the specified order:

```
trident forge -d MyPac --forgeFile myOrder.txt -o MyPac2 --ordered --preserve
```

3. Apply `trident rectify` to increment the package version number and document the reordering:

```
trident rectify -d MyPac2 --packageVersion Minor \  
--logText "reordered the samples alphabetically by Poseidon_ID"
```

`MyPac2` then acts as a stand-in replacement for `MyPac` that only differs in the order of samples (and maybe the order of variables/fields in the `POSEIDON.yml`, `.janno`, `.ssf` or `.bib` files).

5.3 Treatment of the genotype data while merging

Forge performs a series of steps to merge the genotype data of multiple source files:

1. Genotype data from each package is streamed in parallel. Because our packages may have different SNP locations (specified by chromosome-position pairs) listed in their `.bim/.snp` or `.vcf` file, we first perform a zipping-operation, whose behaviour depends on whether `--intersect` is set or not. Without `--intersect`, any SNP position listed in any package will be forwarded to the output, with missing values being filled in in all packages that do not list that particular SNP. With `--intersect`, only SNP positions that are present in all packages are considered. Note that relevant for this step is only whether a given SNP position is part of the genotype data, not whether the actual genotypes are missing or not.
2. At each SNP, the consensus alleles are selected, by collecting all reference and alternative alleles from all sources. If more than two non-dummy alleles (alleles different from `N`) are present in that collection, an error is thrown. If exactly two non-dummy alleles are present (which should be the case for binary SNPs), the two alleles are declared “reference” and “alternative” alleles for the output. If only one non-dummy allele is present, it is set to be the reference allele, and “`N`” is set to be the alternative.
3. All source genotype data is then read and recoded in terms of the two chosen consensus alleles. This will make sure that source data with flipped reference and alternative allele gets correctly merged in.
4. SNP IDs, as part of PLINK `.bim` and `.vcf` files are checked across the source files. If all SNP IDs for a given SNP are missing, then the result will also be missing. If there is only one SNP ID present in some or all source packages, that ID gets forwarded to the output. In the (unusual) case that there are multiple different non-missing SNP IDs (of the form “`rs`” followed by a number), then a debug warning is output (which gets printed to the screen when `--debug` is selected), and simply the first value is chosen to be output into the forged `.bim` file. We decided not to throw an error in that case, because we consider the physical position of the SNP (specified by Chromosome and position) to be definitive, and the SNP ID to be of secondary importance.
5. Genetic positions, as part of PLINK `.bim` files are checked in a similar manner, with “`0.0`” being interpreted as missing.

5.3.1 Strand flips

By default, if incongruent pairs of SNPs are found, an error is thrown, with a message reporting the offending SNPs in all involved packages. The `--strandCheck` option allows `trident` to detect and automatically correct strand flips. Even with `--strandCheck` on there may be cases of pairs that are still un-mergeable, such as `A/T` and `A/G`. `--skipIncongruentSNPs` allows to remove any such SNPs from the output.

259 5.4 Treatment of the .janno file while merging

260 **forge** merges and subsets .janno files along with the genotype data. If a package lacks a .janno file, then a
 261 basic one will be created internally an on-the-fly based on the information in the genotype data, and used for
 262 the output. Missing columns across packages will be filled with n/a.

263 For merging two .janno files **A** and **B** the following rules apply regarding undefined, arbitrary additional
 264 columns:

- 265 • If **A** has an additional column which is not in **B** then empty cells in the rows imported from **B** are filled
 266 with n/a.
- 267 • If **A** and **B** share additional columns with identical column name, then they are treated as semantically
 268 identical units and merged accordingly.
- 269 • In the resulting .janno file, all additional columns from both **A** and **B** are sorted alphabetically and
 270 appended after the normal, specified variables.

271 The following example illustrates the described behaviour:

272 A.janno

Poseidon_ID	Group_Name	Genetic_Sex	AdditionalColumn1	AdditionalColumn2
XXX011	POP1	M	A	D
XXX012	POP2	F	B	E
XXX013	POP1	M	C	F

273 B.janno

Poseidon_ID	Group_Name	Genetic_Sex	AdditionalColumn3	AdditionalColumn2
YYY022	POP5	F	G	J
YYY023	POP5	F	H	K
YYY024	POP5	M	I	L

274 A.janno + B.janno

Poseidon_ID	Group_Name	Genetic_Sex	AdditionalColumn1	AdditionalColumn2	AdditionalColumn3
XXX011	POP1	M	A	D	n/a
XXX012	POP2	F	B	E	n/a
XXX013	POP1	M	C	F	n/a
YYY022	POP5	F	n/a	J	G
YYY023	POP5	F	n/a	K	H
YYY024	POP5	M	n/a	L	I

275 5.5 Treatment of the .ssf file while merging

276 The Sequencing Source File (short .ssf file) is forged in exactly the same way as the .janno file. .ssf files
 277 that are present are included in the forge product, following selection of those entities which are listed in the

poseidon_IDs columns. Columns that are only present in some packages, including those not defined in the Poseidon package specification, are also included in the forged product in the same way as described for .janno files above.

5.6 Treatment of the .bib file while merging

In the forge process all relevant samples for the output package are determined. This includes their .janno entries and therefore the information on the publication keys documented for them in the .janno Publication column. The output .bib file compiles only the relevant references for the samples in the output package. It includes the references exactly once and is sorted alphabetically by key.

5.7 Output modes

The output package of **forge** is created as a new directory **-o**. The title can also be explicitly defined with **-n**. **forge** by default returns a new output package with a generic POSEIDON.yml file, the genotype data as created from the input and the selection, and a .janno file. If the input includes .bib or .ssf files, the output will as well.

Other output formats can be selected with these mutually exclusive flags:

--onlyGeno:

Only the genotype data is returned without any Poseidon package wrapping around it. This is especially useful for data analysis pipelines, where only the genotype data is required.

--minimal:

A minimal output package without .janno, .bib and .ssf. This wraps the genotype data in a very basic Poseidon package.

--preserve:

A full Poseidon package just as the default, but with various settings copied from the source package. This only works in case of a single source package.

For the specific task of sub-setting or reordering (see above) a singular, existing Poseidon package, it can be useful to preserve some fields of the POSEIDON.yml file of this input package, as well as the supplementary information in the README.md and the CHANGELOG.md file. These are typically discarded by **forge**, but can be copied over to the output package with **--preserve**.

--preserve specifically copies the following POSEIDON.yml fields:

- **description**
- **contributor**
- **packageVersion**
- **lastModified**
- **readmeFile**
- **changelogFile**

This does not include the package **title**, which can be easily set to be identical to the source with **-n** or **-o** if it is desired. The **poseidonVersion** field is also not copied, because **trident** can only ever produce output packages with the latest Poseidon schema version.

315 To keep the diff between in- and output package minimal, `--preserve` finally also causes `trident` to keep the
 316 input's order of columns in the `.janno` file, and maintain the input's order of bibtex entries in the `.bib` file.

317 `-z|--zip`

318 With `-z` the genotype data output (independent of the selected output mode) can be wrapped in gzipped archives
 319 with the additional file extension `.gz`. `trident` can seamlessly interact with genotype data in this format.

320 5.8 Other options

321 `forge` has a an optional flag `--intersect`, that defines, if the genotype data from different packages should be
 322 merged with a union or an intersect operation. See *Treatment of the genotype data while merging* above.

323 `--intersect` also influences the automatic determination of the `snpSet` field in the `POSEIDON.yml` file for the
 324 resulting package. If the `snpSets` of all input packages are identical, then the resulting package will just inherit
 325 this configuration. Otherwise `forge` applies the following pairwise merging logic:

Input snpSet A	Input snpSet B	<code>--intersect</code>	Ouput snpSet
Other	*	*	Other
1240K	HumanOrigins	True	HumanOrigins
1240K	HumanOrigins	False	1240K

326 `--selectSnps` allows to provide `forge` with a SNP file in EIGENSTRAT (`.snp`) or PLINK (`.bim`) format to
 327 create a package with a specific selection. When this option is set, the output package will have exactly the
 328 SNPs listed in this file. Any SNP not listed in the file will be excluded. If `--intersect` is also set, only the
 329 SNPs overlapping between the SNP file and the forged packages are output.

330 With `--packagewise` the within-package selection step in `forge` can be skipped. This will result in outputting
 331 all individuals in the relevant packages, and hence a superset of the requested individuals/groups. It may result
 332 in better performance in cases where one wants to forge entire packages.

333 Forged packages can optionally contain a `.janno` column `Source_Package`, which records which original package
 334 (and version) each sample originally came from. This makes it possible to trace a sample back to its origin even
 335 after it has passed through one or more rounds of `forge`. This tracking mechanism is opt-in and can be activated
 336 with the `--addTrace` option.

337 6 Genoconvert command

338 `genoconvert` converts the genotype data in a Poseidon package to a different file format. The respective entries
 339 in the `POSEIDON.yml` file are changed accordingly.

340 Command line details

```
Usage: trident genoconvert ((-d|--baseDir DIR) |
                           ((-p|--genoOne FILE) | --genoFile FILE
                           --snpFile FILE --indFile FILE |
                           --bedFile FILE --bimFile FILE --famFile FILE |
                           --vcfFile FILE) [--snpSet SET])
                           --outFormat FORMAT [-o|--outPackagePath DIR]
```

```
[--removeOld] [--outPlinkPopName MODE] [--onlyLatest]
[-z|--zip]
```

Convert the genotype data in a Poseidon package to a different file format

Available options:

<code>-h,--help</code>	Show this help text
<code>-d,--baseDir DIR</code>	A base directory to search for Poseidon packages.
<code>-p,--genoOne FILE</code>	One of the input genotype data files. Expects .bed, .bed.gz, .bim, .bim.gz or .fam for PLINK, .geno, .geno.gz, .snp, .snp.gz or .ind for EIGENSTRAT, or.vcf or .vcf.gz for VCF. In case of EIGENSTRAT and PLINK, the two other files must be in the same directory and must have the same base name. If a gzipped file is given, it is assumed that the file pairs (.geno.gz, .snp.gz) or (.bim.gz, .bed.gz) are both zipped, but not the .fam or .ind file. If a .ind or .fam file is given, it is assumed that none of the file triples is zipped.
<code>--genoFile FILE</code>	Eigenstrat genotype matrix, optionally gzipped. Accepted file endings are .geno, .geno.gz
<code>--snpFile FILE</code>	Eigenstrat snp positions file, optionally gzipped. Accepted file endings are .snp, .snp.gz
<code>--indFile FILE</code>	Eigenstrat individual file. Accepted file endings are .ind
<code>--bedFile FILE</code>	Plink genotype matrix, optionally gzipped. Accepted file endings are .bed, .bed.gz
<code>--bimFile FILE</code>	Plink snp positions file, optionally gzipped. Accepted file endings are .bim, .bim.gz
<code>--famFile FILE</code>	Plink individual file. Accepted file endings are .fam
<code>--vcfFile FILE</code>	VCF (Variant Call Format) file, optionall gzipped. Accepted file endings are .vcf, .vcf.gz
<code>--snpSet SET</code>	The snpSet of the package: 1240K, HumanOrigins or Other. Only relevant for data input with -p --genoOne or --genoFile + --snpFile + --indFile, because the packages in a -d --baseDir already have this information in their respective POSEIDON.yml files. (default: Other)
<code>--outFormat FORMAT</code>	the format of the output genotype data: EIGENSTRAT, PLINK or VCF.
<code>-o,--outPackagePath DIR</code>	Path for the converted genotype files to be written to. If a path is provided, only the converted genotype files are written out, with no change of the original package. If no path is provided, genotype files will be converted in-place, including a change

	in the POSEIDON.yml file to yield an updated valid package (default: Nothing)
<code>--removeOld</code>	Remove the old genotype files when creating the new ones.
<code>--outPlinkPopName MODE</code>	Where to write the population/group name into the FAM file in Plink-format. Three options are possible: asFamily (default) asPhenotype asBoth. See also <code>--inPlinkPopName</code> .
<code>--onlyLatest</code>	Consider only the latest versions of packages, or the groups and individuals within the latest versions of packages, respectively.
<code>-z,--zip</code>	Should the resulting genotype- and snp-files be gzipped?

341 With the default setting

```
trident genoconvert -d ... -d ... --outFormat EIGENSTRAT|PLINK
```

342 all packages in `-d` will be converted to the desired `--outFormat` (either EIGENSTRAT or PLINK), if the data is
343 not already in this format. This includes updating the respective POSEIDON.yml files.

344 The “old” data is not deleted, but kept around. That means conversion can result in a package with both PLINK
345 and EIGENSTRAT data, but only one is linked in the POSEIDON.yml file, and that is what will be used by
346 `trident`. To delete the old data in the conversion you can add the `--removeOld` flag.

347 `-p` (+ `--snpSet`), `--genoFile` + `--snpFile` + `--indFile` (+ `--snpSet`) (for EIGENSTRAT data),
348 `--bedFile` + `--bimFile` + `--famFile` (+ `--snpSet`) (for PLINK data) or `--vcfFile` (+ `--snpSet`) (for
349 VCF data) allow to directly convert genotype data that is not wrapped in a Poseidon package and store it to a
350 directory given in `-o`. See this example:

```
trident genoconvert \
  -p 2018_Mittnik_Baltic/Mittnik_Baltic.bed \
  --outFormat EIGENSTRAT \
  -o my_directory
```

351 With `-z|--zip` the genotype data output can be wrapped in gzipped archives with the additional file extension
352 `.gz`.

353 7 Jannocoalesce command

354 `jannocoalesce` merges information from one or multiple source `.janno` files into a target `.janno` file.

355 Command line details

```
Usage: trident jannocoalesce ([--pvSource VERSION] (-s|--sourceFile FILE) |
                             (-d|--baseDir DIR)) [--pvTarget VERSION]
                             (-t|--targetFile FILE) (-o|--outFile FILE)
                             [--includeColumns ARG | --excludeColumns ARG]
                             [-f|--force] [--sourceKey ARG] [--targetKey ARG]
                             [--stripIdRegex ARG]
```

Coalesce information from one or multiple janno files to another one

Available options:

<code>-h,--help</code>	Show this help text
<code>--pvSource VERSION</code>	Poseidon version (e.g. 2.7.1). (default: 3.0.0)
<code>-s,--sourceFile FILE</code>	The source .janno file.
<code>-d,--baseDir DIR</code>	A base directory to search for Poseidon packages.
<code>--pvTarget VERSION</code>	Poseidon version (e.g. 2.7.1). (default: 3.0.0)
<code>-t,--targetFile FILE</code>	The target .janno file to fill.
<code>-o,--outFile FILE</code>	File path to write the result to. Can be identical to <code>--targetFile</code> to overwrite the target file in place. Note that trident only writes .janno files in the latest Poseidon version it supports, so in this case v3.0.0.
<code>--includeColumns ARG</code>	A comma-separated list of .janno column names to coalesce. If not specified, all columns that can be found in the source and target will get filled.
<code>--excludeColumns ARG</code>	A comma-separated list of .janno column names NOT to coalesce. All columns that can be found in the source and target will get filled, except the ones listed here.
<code>-f,--force</code>	With this option, potential non-missing content in target columns gets overridden with non-missing content in source columns. By default, only missing data gets filled-in.
<code>--sourceKey ARG</code>	The .janno column to use as the source key. (default: "Poseidon_ID")
<code>--targetKey ARG</code>	The .janno column to use as the target key. (default: "Poseidon_ID")
<code>--stripIdRegex ARG</code>	An optional regular expression to identify parts of the IDs to strip before matching between source and target. Uses POSIX Extended regular expressions.

356 A most basic run may just include two arguments:

```
trident jannocoalesce \  
  --sourceFile path/to/source.janno \  
  --targetFile path/to/target.janno \  
  --outFile path/to/coalesced.janno
```

357 jannocoalesce generally works by reading a source .janno file with `-s|--sourceFile` (or all .janno files in a
358 `-d|--baseDir`) and a target .janno file with `-t|--targetFile`.

359 It then merges these files by a key column, which can be selected with `--sourceKey` and `--targetKey`. The
360 default for both of these key columns is the `Poseidon_ID`. In case the entries in the key columns slightly and
361 systematically differ, e.g. because the `Poseidon_ID`s in either have a special suffix (for example `_SG`), then the

362 `--stripIdRegex` option allows to strip these with a regular expression to thus match the keys.

363 `jannocoalesce` generally attempts to fill **all** empty cells in the target `.janno` file with information from the

364 source. `--includeColumns` and `--excludeColumns` allow to select specific columns for which this should be

365 done. In some cases it may be desirable to not just fill empty fields in the target, but overwrite the information

366 already there with the `-f|--force` option. If the target file should be preserved, then the output can be directed

367 to a new output `.janno` file with `-o|--outFile`.

368 Note that all three files, the source, the target, and the outfile, are mandatory. The roles are:

- 369 • `targetFile` -> This is the file which is taken as the starting point for the new janno file.
- 370 • `sourceFile` -> This is the file from which to read additional columns that might be missing in the target.
- 371 • `outFile` -> This is the file that will contain the coalesced result.

372 In addition to these three files, you can choose a Poseidon Version for both the source (`--pvSource`) and the

373 target (`--pvTarget`), which is useful for backwards compatibility, in case you have `.janno` files in older Poseidon

374 versions. **But:** Note that the output file will *always* be written in the newest Poseidon version. Note that you

375 *can* choose the `outFile` to be the `targetFile`, which will then overwrite the target file. This can be useful if you

376 know what you are doing. Otherwise it is of course safer to choose a new filename for the output.

377 8 Rectify command

378 `rectify` automatically harmonizes POSEIDON.yml files of one or multiple packages. This is not an automatic

379 update from one Poseidon version to the next, but rather a clean-up wizard after manual modifications. It also

380 includes additional, automatic package editing features.

381 Command line details

```
Usage: trident rectify (-d|--baseDir DIR) [--ignorePoseidonVersion]
      [--poseidonVersion ?.??.?]
      [--packageVersion VPART [--logText STRING]]
      [--checksumAll | [--checksumGeno] [--checksumJanno]
      [--checksumSSF] [--checksumBib]]
      [--newContributors DSL] [--jannoRemoveEmpty]
      [--onlyLatest]
```

Adjust POSEIDON.yml files automatically to package changes

Available options:

<code>-h,--help</code>	Show this help text
<code>-d,--baseDir DIR</code>	A base directory to search for Poseidon packages.
<code>--ignorePoseidonVersion</code>	Read packages even if their poseidonVersion is not compatible with trident.
<code>--poseidonVersion ?.??.?</code>	Poseidon version the packages should be updated to: e.g. "2.5.3".
<code>--packageVersion VPART</code>	Part of the package version number in the POSEIDON.yml file that should be updated: Major, Minor or Patch (see https://semver.org).
<code>--logText STRING</code>	Log text for this version in the CHANGELOG file.

<code>--checksumAll</code>	Update all checksums.
<code>--checksumGeno</code>	Update genotype data checksums.
<code>--checksumJanno</code>	Update .janno file checksum.
<code>--checksumSSF</code>	Update .ssf file checksum
<code>--checksumBib</code>	Update .bib file checksum.
<code>--newContributors DSL</code>	Contributors to add to the POSEIDON.yml file in the form "[Firstname Lastname](Email address)<ORCID>;...". The ORCIDs are optional.
<code>--jannoRemoveEmpty</code>	Reorder the .janno file and remove empty columns. Remember to pair this option with <code>--checksumJanno</code> to also update the checksum.
<code>--onlyLatest</code>	Consider only the latest versions of packages, or the groups and individuals within the latest versions of packages, respectively.

382 It can be called with a lot of optional arguments. Note that `rectify` by default does **not** apply any changes if
 383 none of these arguments are set. Each change requires explicit opt-in.

```
trident rectify -d ... -d ... \
  --poseidonVersion "X.X.X" \
  --packageVersion Major|Minor|Patch \
  --logText "short description of the update" \
  --checksumAll \
  --newContributors "[Firstname Lastname](Email address)<ORCID>;..." \
  --jannoRemoveEmpty
```

384 The following arguments determine which fields of the POSEIDON.yml file should be modified:

- 385 • `--poseidonVersion` allows a simple change of the `poseidonVersion` field in the POSEIDON.yml file.
- 386 • `--packageVersion` increments the package version number in the first, the second or the third position.
 387 It can optionally be called with `--logText`, which appends an entry to the CHANGELOG file for the
 388 respective package version update. `--logText` also creates a new CHANGELOG.md file if it does not exist
 389 yet.
- 390 • `--checksumGeno`, `--checksumJanno`, `--checksumSSF` and `--checksumBib` add or modify the respective
 391 checksum fields in the POSEIDON.yml file. `--checksumAll` is a wrapper to call all of them at once.
- 392 • `--newContributors` adds new contributors.

393 As `rectify` reads and rewrites POSEIDON.yml files, it may change their inner order, layout or even content (e.g. if
 394 they have fields which are not in the POSEIDON.yml specification). Create a backup of the POSEIDON.yml file
 395 before running `rectify` if you are uncertain if this might affect you negatively.

396 `--jannoRemoveEmpty` is the first option that does not edit POSEIDON.yml, but .janno files. It allows to remove
 397 empty columns from .janno files, so columns that only feature empty strings or `n/a` values. As part of this process
 398 it reorders the columns of the .janno file. Remember to pair this option with `--checksumJanno` or `checksumAll`
 399 to automatically update the .janno checksum in the POSEIDON.yml file afterwards.

400 9 List command

401 `list` lists packages, groups, individuals and bibliography entries of local Poseidon package datasets, or of
402 packages available in the archives on the web server.

403 Command line details

```
Usage: trident list ((-d|--baseDir DIR) | --remote [--remoteURL URL]
                    [--archive STRING])
                    (--packages [--fullOutput] | --groups | --individuals
                    [--fullJanno | [-j|--jannoColumn COLNAME]] |
                    --bibliography [--fullBib | [-b|--bibField BIB-FIELD]])
                    [--raw] [--onlyLatest]
```

List packages, groups or individuals from local or remote Poseidon repositories

Available options:

<code>-h,--help</code>	Show this help text
<code>-d,--baseDir DIR</code>	A base directory to search for Poseidon packages.
<code>--remote</code>	List packages from a remote server instead the local file system.
<code>--remoteURL URL</code>	URL of the remote Poseidon server. (default: "https://server.poseidon-adna.org")
<code>--archive STRING</code>	The name of the Poseidon package archive that should be queried. If not given, then the query falls back to the default archive of the server selected with <code>--remoteURL</code> . See the archive documentation at https://www.poseidon-adna.org/#/archive_overview for a list of archives currently available from the official Poseidon Web API. (default: Nothing)
<code>--packages</code>	List all packages.
<code>--fullOutput</code>	extend the output to include information contained the POSEIDON.yml file
<code>--groups</code>	List all groups, ignoring any group names after the first as specified in the .janno-file.
<code>--individuals</code>	List all individuals/samples.
<code>--fullJanno</code>	output all Janno Columns
<code>-j,--jannoColumn COLNAME</code>	List additional fields from the janno files, using the .janno column heading name, such as "Country", "Site", "Date_C14_Uncal_BP", etc... Can be given multiple times
<code>--bibliography</code>	output bibliography information for packages
<code>--fullBib</code>	output all bibliography fields found in any bibliography item
<code>-b,--bibField BIB-FIELD</code>	List information from the given bibliography field, for example "abstract" or "publisher". Can be given

	multiple times.
<code>--raw</code>	Return the output table as tab-separated values without header. This is useful for piping into <code>grep</code> or <code>awk</code> .
<code>--onlyLatest</code>	Consider only the latest versions of packages, or the groups and individuals within the latest versions of packages, respectively.

404 To list packages from your local repositories, as seen above you can run

```
trident list -d ... -d ... --packages
```

405 This will yield a nicely formatted table of all packages, their version and the number of individuals in them.

406 With `--fullOutput` the table includes additional fields from the packages' POSEIDON.yml files.

407 You can use `--remote` to show packages on the remote server. For example

```
trident list --packages --remote --archive "community-archive"
```

408 will result in a view of all packages available in one of the public Poseidon archives. Just as for `fetch`, the

409 `--archive` flag allows to choose which public archive to query.

410 Independent of whether you query a local or an online archive, you can not just list packages, but also groups,

411 as defined in the third column of EIGENSTRAT .ind files (or the first/last column of a PLINK .fam file),

412 and individuals with the flags `--groups` and `--individuals` (instead of `--packages`). `--bibliography` returns

413 publication-wise bibliography information.

414 The `--individuals` flag additionally provides a way to immediately access information from .janno files

415 on the command line. This works with the `-j|--jannoColumn` option. For example adding `-j Country -j`

416 `Date_C14_Uncal_BP` to the commands above will add the `Country` and the `Date_C14_Uncal_BP` columns to the

417 respective output tables. `--fullJanno` outputs all available columns.

418 Analogously, with `--bibliography` additional fields from the .bib files can be added to the output table with

419 `-b|--bibField ...` and `--fullBib`. `-b journal`, for example, adds a column with the publication's journal.

420 Note that if you want a less ornate table, for example because you want to load this into Excel, or pipe into

421 another command that cannot deal with the table layout, you can use the `--raw` option to output that table as

422 a simple tab-delimited stream.

423 10 Summarise command

424 `summarise` prints some general summary statistics for a given Poseidon dataset taken from the .janno files.

425 Command line details

```
Usage: trident summarise (-d|--baseDir DIR) [--raw]
```

Get an overview over the content of one or multiple Poseidon packages

Available options:

<code>-h,--help</code>	Show this help text
------------------------	---------------------

<code>-d,--baseDir DIR</code>	A base directory to search for Poseidon packages.
<code>--raw</code>	Return the output table as tab-separated values without header. This is useful for piping into <code>grep</code> or <code>awk</code> .

426 You can run it with

```
427 trident summarise -d ... -d ...
```

427 which will show you context information like – among others – the number of individuals in the dataset, their sex
 428 distribution, the mean age of the samples or the mean coverage on the 1240K SNP array in a table. **summarise**
 429 depends on complete `.janno` files and will silently ignore missing information.

430 You can use the `--raw` option to output the summary table in a simple, tab-delimited layout.

431 11 Survey command

432 **survey** tries to indicate package completeness (mostly focused on `.janno` files) for Poseidon datasets.

433 Command line details

```
Usage: trident survey (-d|--baseDir DIR) [--raw] [--onlyLatest]
```

Survey the degree of context information completeness for Poseidon packages

Available options:

<code>-h,--help</code>	Show this help text
<code>-d,--baseDir DIR</code>	A base directory to search for Poseidon packages.
<code>--raw</code>	Return the output table as tab-separated values without header. This is useful for piping into <code>grep</code> or <code>awk</code> .
<code>--onlyLatest</code>	Consider only the latest versions of packages, or the groups and individuals within the latest versions of packages, respectively.

434 Running

```
trident survey -d ... -d ...
```

435 will yield a table with one row for each package. See `trident survey -h` for a legend which cell of this table
 436 means what.

437 Again you can use the `--raw` option to output the survey table in a tab-delimited format.

438 12 Validate command

439 **validate** checks Poseidon packages and individual package components for structural correctness.

440 Command line details

```
Usage: trident validate ((-d|--baseDir DIR) [--ignoreGeno] [--fullGeno]
                        [--forgeTest] [--strandCheck] [--ignoreDuplicates]
                        [-c|--ignoreChecksums] [--ignorePoseidonVersion] |
                        --pym1 FILE | (-p|--genoOne FILE) | --genoFile FILE
                        --snpFile FILE --indFile FILE |
                        --bedFile FILE --bimFile FILE --famFile FILE |
                        --vcfFile FILE | [--pvJanno VERSION] --janno FILE |
                        [--pvSSF VERSION] --ssf FILE |
                        --bib FILE) [-j|--mandatoryJannoColumn COLNAME]
                        [-s|--mandatorySSFColumn COLNAME] [--noExitCode]
                        [--onlyLatest]
```

Check Poseidon packages or package components for structural correctness

Available options:

<code>-h,--help</code>	Show this help text
<code>-d,--baseDir DIR</code>	A base directory to search for Poseidon packages.
<code>--ignoreGeno</code>	Ignore snp and geno file.
<code>--fullGeno</code>	Test parsing of all SNPs in each package (by default only the first 100 SNPs are probed).
<code>--forgeTest</code>	Even more extensive than <code>--fullGeno</code> : Test forging the entire dataset. This will detect incongruent SNPs between the packages. Can be called with <code>--strandCheck</code> .
<code>--strandCheck</code>	Whether to allow strand flips in the genotype data. Note that this will remove any A/T and G/C SNPs from the data, as for those we cannot determine the correct strand orientation.
<code>--ignoreDuplicates</code>	Do not stop on duplicated individual names in the package collection.
<code>-c,--ignoreChecksums</code>	Whether to ignore checksums. Useful for speedup in debugging.
<code>--ignorePoseidonVersion</code>	Read packages even if their poseidonVersion is not compatible with trident.
<code>--pym1 FILE</code>	Path to a POSEIDON.yml file.
<code>-p,--genoOne FILE</code>	One of the input genotype data files. Expects .bed, .bed.gz, .bim, .bim.gz or .fam for PLINK, .geno, .geno.gz, .snp, .snp.gz or .ind for EIGENSTRAT, or.vcf or .vcf.gz for VCF. In case of EIGENSTRAT and PLINK, the two other files must be in the same directory and must have the same base name. If a gzipped file is given, it is assumed that the file pairs (.geno.gz, .snp.gz) or (.bim.gz, .bed.gz) are both zipped, but not the .fam or .ind file. If a .ind or .fam file is given, it is assumed that none of the

```

file triples is zipped.
--genoFile FILE      Eigenstrat genotype matrix, optionally gzipped.
                      Accepted file endings are .geno, .geno.gz
--snpFile FILE       Eigenstrat snp positions file, optionally gzipped.
                      Accepted file endings are .snp, .snp.gz
--indFile FILE       Eigenstrat individual file. Accepted file endings are
                      .ind
--bedFile FILE       Plink genotype matrix, optionally gzipped. Accepted
                      file endings are .bed, .bed.gz
--bimFile FILE       Plink snp positions file, optionally gzipped.
                      Accepted file endings are .bim, .bim.gz
--famFile FILE       Plink individual file. Accepted file endings are .fam
--vcfFile FILE       VCF (Variant Call Format) file, optionall gzipped.
                      Accepted file endings are .vcf, .vcf.gz
--pvJanno VERSION    Poseidon version (e.g. 2.7.1). (default: 3.0.0)
--janno FILE         Path to a .janno file.
--pvSSF VERSION      Poseidon version (e.g. 2.7.1). (default: 3.0.0)
--ssf FILE           Path to a .ssf file.
--bib FILE           Path to a .bib file.
-j,--mandatoryJannoColumn COLNAME
                      Usually optional .janno file column that should be
                      treated as mandatory, such as e.g. Individual_ID. Can
                      be given multiple times.
-s,--mandatorySSFColumn COLNAME
                      Usually optional .ssf file column that should be
                      treated as mandatory, such as e.g. poseidon_IDs. Can
                      be given multiple times.
--noExitCode         Do not produce an explicit exit code.
--onlyLatest         Consider only the latest versions of packages, or the
                      groups and individuals within the latest versions of
                      packages, respectively.

```

441 You can run it with

```
trident validate -d ... -d ...
```

442 to check packages and it will either report a success (**Validation passed**) or failure with specific error messages.

443 Instead of validating entire packages with **-d** you can also apply it to individual files and package compo-
 444 nents: **--pym1** (POSEIDON.yml), **-p** | **--genoFile** + **--snpFile** + **--indFile** | **--bedFile** + **--bimFile**
 445 + **--famFile** | **--vcfFile** (genotype data), **--janno** (.janno file), **--ssf** (.ssf file) or **--bib** (.bib file). In this
 446 case **validate** attempts to read and parse the respecitve files individually and reports any issues it encounters.
 447 Note that this considers the files in isolation and does not include any cross-file consistency checks.

448 When applied to packages, **validate** tries to ensure that each package adheres to the Poseidon package
 449 specification. Here is a list of what is checked:

- 450 • Structural correctness of the POSEIDON.yml file.

- 451 • Presence of all files references in the POSEIDON.yml file.
- 452 • Full structural correctness of .janno, .ssf and .bib file.
- 453 • Superficial correctness of genotype data files by parsing the first 100 SNPs. A full check that parses all SNPs
- 454 can be triggered with the `--fullGeno` option. The `--forgeTest` option goes even further and simulates
- 455 a full merge of all data across packages to detect incongruent SNPs. `--ignoreGeno`, on the other hand,
- 456 causes `validate` to ignore the genotype data entirely, which speeds up the validation significantly.
- 457 • Correspondence of BibTeX keys in .bib and .janno
- 458 • Correspondence of sample IDs in .janno and .ssf.
- 459 • Correspondence of sample and group IDs in .janno and genotype data files.

460 In fact much of this validation already runs as part of the general package reading pipeline invoked for other
461 `trident` subcommands (e.g. `forge`). `validate` is meant to be more thorough and brittle, though, and will
462 explicitly fail if even a single package is broken. For special cases more flexibility can be enabled with the options
463 `--ignoreDuplicates`, `--ignoreChecksums` and `--ignorePoseidonVersion`.

464 Remember to run `validate` with `--debug` to get more information in case the default output is not sufficient to
465 analyse an issue.